AI-Augmented Project Proposal Methodology

Engineering Business Value into Information Systems

1 Motivation: Why Are We Doing This?

In modern Software Engineering, the most expensive mistake is building the *wrong* thing efficiently. This guide introduces a systematic method for using Generative AI as a *Process Analysis Specialist*. You will use the AI to stress-test your architecture and business logic before drafting your proposal. But more importantly, you are doing this to build a **reusable context engine** for the rest of your semester.

By defining your project through structured, machine-readable artifacts now, you are creating the exact input files you will feed to coding agents during the implementation phase. Good artifacts prevent "context rot" later, ensuring your AI coding assistants actually understand your business logic and system boundaries.

Read this document completely before starting.
Use at most 10 work hours to create the Proposal Deliverables.

2 The Artifact-Driven Methodology

Engineering a complex Information System requires accurate context. Unstructured AI chat histories quickly turn into hallucinated requirements and overloaded feature lists. To avoid this, you will use an **Artifact-Driven** approach: you will create concise, plain-text files (Artifacts) as your ground truth. You will then instruct the AI to review, critique, and challenge these files.

Note: You are completely free to expand on this template. We encourage you to use advanced agentic setups, custom Python scripts, or specialized models to reach your conclusions, provided you maintain the core principles of artifact-driven context and critical human oversight.

3 What You Actually Need to Hand In

Your final submission should be straightforward and functional:

- **The Project Proposal Document:** A cohesive PDF covering the core areas mentioned in Section 5. Keep it focused and impactful.
- **The Machine-Readable Artifacts:** The Markdown files from Section 5.3 that prove you have a solid, prompt-ready foundation for the rest of the project.
- **AI Session Links:** Persistent links, exported transcripts, or code repositories (if using custom agent scripts) of the AI sessions used during your brainstorming. Please include your used AI Model.
- **Self-Reflection Log:** Your answers to the three critical reflection questions from Phase 4.

4 The Value Proposition Framework

The core of your project is its business logic. Your objective during the AI sparring phase is to find answers to the following points. The AI must validate your project against this specific framework:

(1) Sponsor: Who will decide to provide the resources and data to complete your project?

(2.1) Pain Point of the Sponsor: What pain point shall your project result address?

(2.2a) Process of Interest: Which specific process fails to provide a desired outcome?

(2.2b) Undesired Process Outcome: What process outcome has insufficient quality/quantity?

(2.3) Undesired Condition: What condition will you improve to address the pain point?

(2.4) Evaluation: How can you or the customer measure this improvement?

(3) Product/Service: What product/service functions do you propose to deliver?

(4.1) Impact: What is the business impact of the product or service for the sponsor?

(4.2) Comparison: How does this compare to alternative solution approaches?

(5) Root Causes: What likely causes of the undesired condition could you improve?

(6) Strategic Promise: Why is your approach a good investment for your sponsor?

5 Workflow: From Context to Artifacts

Follow this 4-phase process to ensure you maintain executive control over the AI's contributions.

5.1 Phase 1: Human-Led Context Bootstrapping

Before opening any AI tool, sit down with your team and draft a short, plain-text **Domain Brief**. This establishes your project's reality. It needs to cover at least:

- **Project Background & Motivation:** The real-world inspiration, domain context, or sponsor contact driving the project.
- **Technical & Resource Constraints:** The tech stack your team actually knows and is aiming to work with, your time resources, and explicit out-of-scope boundaries.
- **Proposed Solution Sketch:** A basic outline of the problem and your envisioned software architecture.

The output of all subsequent steps will be better and faster, the more precise you are here.

5.2 Phase 2: Value Proposition Sparring

Feed your Domain Brief and the Value Proposition Framework (Section 4) into your chosen AI. Instruct it to act as a strict Process Analyst. To ensure the AI strictly challenges your assumptions rather than just rewriting them, we highly recommend using a structured prompt. You can find an exemplary, ready-to-copy instruction prompt in **Appendix A**.

Debate in your team and with the AI. Reject fluff, correct its misunderstandings of your constraints, and refine your business logic until the 11 framework points are solidly answered.

5.3 Phase 3: Agent-Ready Artifact Generation

Structure the validated decisions into two Markdown files. Keep them clean and concise, as they are the context files you might want to feed to AI coding agents later in the semester:

- `business_logic.md`: The finalized 11-point Value Proposition Framework and stakeholder win-conditions from Section 5.2.
- `domain_architecture.md`: Core entities, system boundaries, and your chosen tech stack. We encourage you to use a text-based format (e.g., PlantUML) in preparation for your implementation phase.

5.4 Phase 4: AI Self-Reflection

Self-reflection is not just administrative paperwork for this course; it is a crucial engineering practice. Actively questioning the AI prevents you from losing your connection to the project and ensures you remain the primary domain expert. Document your reflections on the following points:

Self-Reflection Log

1. The Reality Check: Where did the AI oversimplify the technical, social, or market realities of your project, and how did you correct its assumptions?
2. Resource Alignment: Give a specific example of how you had to scale back the AI's overly optimistic feature suggestions to realistically fit your team's constraints.
3. Creative Ownership: Describe an instance where your team's local knowledge, technical experience, or domain expertise overrode a generic AI suggestion to produce a better architectural or business decision.

6 Collaborating with AI on the Final Proposal

Your final Project Proposal should be a cohesive narrative built from your artifacts. We encourage the **intelligent usage of AI** to help draft, structure, and polish the document. You can feed your three markdown artifacts to the AI and have it generate sections based on your validated logic.

While we provide a baseline structure, we are primarily interested in the depth of your business logic and how well you address the core critique points. Feel free to adapt the structure to best fit your narrative, as long as it cleanly covers the following core areas:

- **Project Overview:** What is it, who is it for, the team structure, and the core goals/milestones.
- **Initial Situation:** The current broken business process, its limitations, and the justification for a software engineering intervention.
- **The Proposed Solution:** Your domain description (including a visual UML Diagram), planned features, and the core findings from your Value Proposition (Section 4).

A Exemplary Instruction Prompt

Copy and paste the following Markdown text into your chosen LLM alongside your project context. The AI will act as a sparring partner to help you fill out the Value Proposition framework.

Sparring Partner Prompt (Markdown)

```
# Role
You are an experienced Process Analyst supporting a student team in drafting a Value
     Proposition for a software project. You are a critical but constructive
    sparring partner: you help the team discover flaws, sharpen vague ideas, and
    make well-founded decisions. The team makes the final call on every answer.

# Goal
Together, we will work out a **Value Proposition** by answering the questions listed
     below. The final result is a Markdown document that will later serve as context
     for AI-assisted coding, so it must be precise, concrete, and free of marketing
    fluff.

# How we work (one question at a time)
For each question, follow this loop:
1. **Draft:** Propose a concrete answer (2-5 sentences) based on our project idea
    and everything agreed so far. Do not hold back a solution. Always give your best
     draft.
2. **Expose assumptions:** Briefly list the assumptions your draft relies on,
    especially ones we have not confirmed.
3. **Challenge:** Ask at most 2-3 focused questions that test the weakest point of
    the draft. Prefer questions that make us think about evidence, measurability,
    and feasibility.
4. **Iterate:** When we respond, revise the draft and say briefly what changed and
    why. If our answer is vague or contradicts earlier answers, say so directly and
    politely.
5. **Lock in:** Only move on when we confirm (e.g., "ok", "next"). Then show the
    final version of the answer and continue with the next question.

Keep each response short and focused. Do not repeat the full document unless we ask
    for it.

# Commands we can use
- ‘next‘: accept the current draft and move on
- ‘skip‘: postpone this question and come back to it later
- ‘deeper‘: challenge the current answer more thoroughly
- ‘summary‘: show the Value Proposition so far as Markdown
- ‘review‘: check all locked answers for contradictions, gaps, and unrealistic scope

# Guiding principles
- Keep the whole proposal consistent: later answers must fit earlier ones. Point out
     when a new answer affects an earlier one.
- Continuously check feasibility against our constraints (time budget, team skills,
    out-of-scope boundaries). Warn us early if the scope is too large.
- Distinguish clearly between the sponsor’s perspective and our (the team’s)
    perspective.
- Favor measurable, specific statements over general ones (e.g., "reduce manual data
     entry from ~4h to <1h per week" instead of "improve efficiency").
- If information is missing that only we can provide, ask for it instead of
    inventing it, and mark any placeholder clearly as ‘[TO BE CONFIRMED]‘.

# Value Proposition Questions
```

```
1. Sponsor: Who will decide to provide the resources and data needed to complete the
     project?
2. Pain Point
   - 2.1 Pain Point of the Sponsor: Which pain point shall the project result
    address?
   - 2.2a Process of Interest: Which specific process fails to deliver a desired
    outcome?
   - 2.2b Undesired Process Outcome: Which process outcome has insufficient quality
    or quantity?
   - 2.3 Undesired Condition: Which condition will you improve to address the pain
    point?
   - 2.4 Evaluation: How can you or the sponsor measure this improvement?
3. Product/Service: Which product or service functions do you propose to deliver?
4. Impact
   - 4.1 Business Impact: What is the business impact of the product or service for
    the sponsor?
   - 4.2 Comparison: How does this compare to alternative solution approaches?
5. Root Causes: Which likely causes of the undesired condition can you influence?
6. Strategic Promise: Why is your approach a good investment for the sponsor?

# Final Output
When all questions are locked in, run a final ‘review‘, then output the complete
    Value Proposition as a clean Markdown document with one heading per question,
    followed by a short section "Open Points & Risks".

# Our Project Idea
**Project Background & Motivation:**
[Real-world inspiration, domain context, or sponsor contact driving the project]

**Technical & Resource Constraints:**
[Tech stack the team actually knows; time budget of 720 / 900 hours; explicit out-of
    -scope boundaries]

**Proposed Solution Sketch:**
[Rough outline of the problem and the envisioned software architecture]

Start by briefly summarizing our project idea in 3-4 sentences and naming the
    biggest uncertainty you see. Then begin with Question 1.
```